REVA UNIVERSITY • SCHOOL OF COMPUTING & INFORMATION TECHNOLOGY

AIML-401 • OFFICIAL FACULTY LECTURE MATERIAL

Course Material: CNN Architectures, Feature Maps & Spatial Convolutions

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1. Primary Teaching Points & Concept Excerpt

“Convolutional Neural Networks (CNNs) are specialized deep learning architectures designed for processing grid-structured topological data such as images. By enforcing parameter sharing across receptive fields and translation equivariance, CNNs extract spatial hierarchies of features ranging from low-level edges to high-level semantic representations.”

2. Mathematical Formulation & Core Equations

1. 2D Discrete Cross-Correlation Operator:

$$S(i, j) = (I * K)(i, j) = \sum_m \sum_n I(i+m, j+n) \cdot K(m, n) + b$$

2. Spatial Dimension Transformation:

$$O = \blacksquare(W - F + 2P) / S\blacksquare + 1$$

where W = input spatial size, F = filter kernel width, P = padding, S = stride.

3. Residual Skip-Connection Identity Mapping:

$$y = F(x, \{W_i\}) + x \implies \partial E / \partial x = (\partial E / \partial y) \cdot (\partial F/\partial x + 1)$$

3. 4-Stage Algorithmic Architecture Pipeline

Stage	Pipeline Component	Mathematical / Architectural Operation	Key Objective
1	Input Layer	Raw tensor ingestion (H x W x C channels)	Standardization [0, 1]
2	Conv2D + ReLU	Parametric kernel convolution with non-linear thresholding	Spatial feature extraction
3	Pooling	Spatial downsampling via Max-Pool (2x2, stride 2)	Translation invariance
4	Dense Head	Flattened tensor passed into Softmax classification layer	Class logit distribution

4. High-Yield Question & Marking Rubric for Midterm IA-2 & SEE Exams

Expected University Exam Question (8 Marks): *'Explain the architecture of a Convolutional Neural Network. Calculate the output feature map size for a 224x224x3 image passed through 64 filters of size 7x7 with stride 2 and padding 3. Explain how ResNet skip connections eliminate the vanishing gradient problem.'*

Official Faculty Marking Rubric:

• 2 Marks: Mathematical formulation of convolution and output size formula.

• 3 Marks: Dimension calculation: O = (224 - 7 + 2*3)/2 + 1 = 112x112x64.

• 3 Marks: Residual skip-connection derivation and gradient preservation proof.